भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

Indian Institute of Information Technology Guwahati

CS301 Theory Of Computation, September 2024

Mid-semester Test

Duration - 2 hours

Question:	1	2	3	4	5	6	Total
Points:	10	5	6	6	8	5	40

Answers without proper justification will fetch zero marks

1. (a) A language over alphabet $\{a, b\}$ is said to be cofinite if it contains all strings over that alphabet except for some finite number of strings. Prove that every cofinite language is regular. [3]

   (b) We can prove that two regular expressions are equivalent by showing that their minimum state Finite Automaton are the same, except for state names. Using this technique, show that the following regular expressions are equivalent: [3]

   1. $(a + b)^*$
   2. $(a^* + b^*)^*$
   3. $((\epsilon + a)b^*)^*$

   (c) Design a minimum state deterministic finite automation accepting all the binary strings of $0's$ & $1's$ which are divisible by 3 but not divisible by 5. [4]

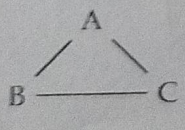
2. Consider the following statement about a language L.

   $$\exists N\ \forall w \in L\ \exists x, y, z : (|w| > N) \Rightarrow ((w = xyz) \wedge (y \neq \epsilon) \wedge (|xy| \leq N) \wedge (\forall i \geq 0 : xy^i z \in L))$$

   (a) What can you say about the class of languages for which it is true? [1]

   (b) Form the logical negation of this statement, using the known behaviour of quantifiers under negation in order to move the negation, $\neg$, from left to right through the statement. [4]

3. Prove that the language of binary string representations of adjacency matrices of graphs is not regular. For example, Figure 1, represents a graph, the corresponding adjacency matrix and string. [6]



	A	B	C
A	0	1	1
B	1	0	1
C	1	1	0

The string corresponding to G is 011101110

Figure 1: Graph G, Adjacency matrix A of G, and string corresponding to A.

4. Use the Pumping Lemma for CFLs to prove that the language $\{a^{n^2} : n \in \mathbb{N}\}$ is not context-free. [6]

5. The language **Luke** has the following Context-Free-Grammar:
Let G be the following CFG.

$$S \to Z$$
$$S \to Sooo$$
$$Z \to nZ$$
$$Z \to \epsilon$$

(a) Construct a diagram to give a Pushdown Automaton for Luke. [3]

(b) Is Luke a regular language? [2]

(c) Convert the grammar into Chomsky Normal Form. [3]

6. Suppose you have a Turing machine M. In this question, we will use the following propositions about M.
For each time t, each tape position i and each symbol s, the proposition $A_{t,i,s}$ means:

At time t, tape cell i contains symbol s.

For each time t and each state q, the proposition $B_{t,q}$ means:

At time t, the machine is in state q.

For each time t and each tape position i, the proposition $C_{t,i}$ means:

At time t the Tape Head is at tape cell i.

The time t can be any non-negative integer (with start time being $t = 0$). The tape cell i can be any positive integer. The state q can be any number in $\{1, 2, \cdots, N\}$, where N is the number of states. The symbol s is a member of $\{a, b, \Delta\}$, where as usual Δ denotes the blank symbol. Use the above propositions to construct logical expressions with the following meanings.

(a) At time t, tape cell i has at least one symbol in it. [1]

(b) At time t, tape cell i has at most one symbol in it. [1]

(c) At time t, tape cell i has exactly one symbol in it. [1]

(d) At time t, and with Tape Head at tape cell i, if the current state is p and the symbol in cell i is x, then at time $t + 1$ the state is q, the symbol in cell i is y, and the Tape Head is at cell $i + d$, where $d \in \{\pm 1\}$. [2]

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